

MUHAMMAD SHAYAN AHMED

ML ENGINEER | MACHINE LEARNING | DEEP LEARNING

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OBJECTIVE

Software Engineering student specializing in Machine Learning, Deep Learning, and Generative AI, with hands-on experience building and deploying end-to-end ML pipelines, computer vision systems, and full-stack AI applications. Seeking an ML/AI Engineering role to apply production-oriented model development, API integration, and deployment skills to real-world systems.

EXPERIENCE

ML Engineering Intern (Remote) — [FlyRank AI](#)

Jun 2026 – Present

- Scoped end-to-end machine learning problem statements, defining decision objectives, unit of analysis, target outputs, and evaluation criteria to align model design with business requirements.
- Performed exploratory data analysis (EDA) in Python and Jupyter Notebook to validate data quality and feasibility, reducing risk in downstream model planning.
- Documented ML workflows, experimentation plans, and research framing in collaboration with mentors, following industry-standard MLOps and reproducibility practices.

Data Science & Analytics Intern (Remote) — [DevelopersHub Corporation](#)

Apr 2026 – Jun 2026

- Analyzed real-world datasets using Python (Pandas, NumPy) and SQL across a 6-week structured program, surfacing actionable business insights from raw data.
- Built statistical models and applied data preprocessing, feature engineering, and EDA to strengthen data-driven decision-making.
- Delivered business reports and data visualizations to stakeholders, improving clarity and adoption of analytical findings across end-to-end data pipelines.

PROJECTS

Aeterna AI – AI Interior Design Platform [\[Live\]](#) [\[GitHub\]](#)

React · FastAPI · SegFormer · MiDaS · Stable Diffusion

- Engineered a production-grade AI interior design platform using React, FastAPI, and PyTorch, transforming user-uploaded room images into professionally redesigned spaces via a full-stack pipeline.
- Designed a custom Preference ANN — a 3-layer MLP with LayerNorm and 5 output heads — encoding room type, style, density, lighting, and past rating signals into a 32-dimensional feature vector to personalize design generation.
- Integrated SegFormer for semantic segmentation, MiDaS for depth estimation, a custom ANN for preference learning, and Stable Diffusion for personalized image generation, combining four ML models into a single inference pipeline.
- Deployed a scalable architecture with backend on Railway and frontend on Vercel, reducing the production bundle from ~900KB to ~200KB via Vite bundle splitting, enabling real-time image processing and API-driven design generation.

AI Medical Diagnostics Platform [\[Live\]](#) [\[GitHub\]](#)

React · Flask · TensorFlow · CNN · OpenCV · Vercel

- Built a full-stack AI diagnostics platform unifying Pneumonia Detection and Skin Cancer Classification into a single React dashboard backed by a Flask REST API.
- Trained and deployed CNN-based deep learning models for chest X-ray and dermoscopic image classification, achieving approximately 90% classification accuracy, with confidence-score visualization to support clinical interpretability.

Interactive Business Dashboard [\[Live\]](#) [\[GitHub\]](#)

Streamlit · Plotly · Pandas

- Developed a BI dashboard in Streamlit and Plotly analyzing sales, profit, and customer performance across a 51,291-row Global Superstore dataset, turning raw transactional data into actionable business insights.
- Implemented dynamic region/category filters and monthly trend analytics, enabling real-time, multi-dimensional exploration of performance metrics.

SKILLS

Programming Languages: Python, SQL, Java, JavaScript

ML / AI Frameworks: Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy

Data Visualization & BI: Power BI, Matplotlib, Seaborn, Excel

Web & APIs: Flask, React.js, FastAPI, Streamlit, Postman, REST APIs

Databases: MySQL, MongoDB, SQL querying, data modeling

Tools & Platforms: Git, GitHub, GitHub Actions (CI/CD), Jupyter Notebook, Google Colab, VS Code, Anaconda

Soft Skills: Analytical Thinking, Problem Solving, Communication, Leadership, Team Collaboration, Adaptability

CERTIFICATIONS

Data Fundamentals — IBM SkillsBuild [\[Certificate\]](#)

Data Analytics Essentials — Cisco Networking Academy [\[Certificate\]](#)

Machine Learning for Data Science Projects — IBM SkillsBuild [\[Certificate\]](#)

EDUCATION

Sir Syed University of Engineering & Technology (SSUET), Karachi

Expected July 2027

Bachelor of Science in Software Engineering — Batch 2023F